Blockchain Forensics Platform: test data sheet

Inputs to type or upload yourself, with the result the system should give. Measured on 11 October 2026.

Before you start

1. Use the **demo platform** at `http://localhost:8502` and sign in as `supervisor`. Several results depend on its watchlist of OFAC-sanctioned addresses, so the normal platform on port 8501 will score them lower. To start it: `.venv/bin/python demo/seed_demo.py`, then `PLATFORM_DB=data/demo.db .venv/bin/streamlit run app.py --server.port 8502`.
2. The wallets are real and the data is live, so scores can drift as the chains move. Wallets that have gone quiet (most of the sanctioned ones) stay stable. Busy exchange wallets change daily.
3. Default bands for wallets: **High** ≥ 50, **Medium** 20–49, **Low** under 20. If you run the feedback-loop test, the thresholds change. Re-run the seed to reset them.
4. Every address is a public example of a *pattern*, such as a sanctions listing, mixing or pass-through. It is not an accusation against its owner. All mobile-money accounts are fake `TEST-ACC-…` placeholders.

1. USDT wallets on TRON

Investigate → **TRON / USDT wallets**. Paste the address and press Investigate.

#	Address	Expected	Why
T1	`TASWbk6X1wiTku5TMmMQYqYFvshVEtfJy8`	**80 High**	OFAC-listed (on the watchlist) plus rapid pass-through
T2	`TPFa9gQW2Mvebjo74FW6ZdFY2GUCg9V9xB`	**70 High**	Dealt directly with a sanctioned wallet, pass-through, P2P-like behaviour
T3	`TEdvb5Rff8fSWEeMZwXCHA8WmZCXRRtiQr`	**65 High**	Exposure to a sanctioned wallet, pass-through, mostly round amounts
T4	`TEEYCuGDyeNkuDj4u6GQRXxXo3Nh29r2vP`	**60 High**	OFAC-listed; about 1.26 M USDT moved
T5	`TXSsAGqvTMBtC5TRV4prdBjdh5ZMD2kwe1`	**45 Medium**	Pass-through, round amounts, new wallet with large volume
T6	`TKCkaxc1hPo7XmHNas6h4GvauHoyBVW8Bm`	**40 Medium**	Exposure to a sanctioned wallet only
T7	`TGFWLzVvyDdwnrN3XKKGsqthFf61nTX155`	**35 Medium**	Pass-through, fan-in (collection wallet)
T8	`TDxQY8axwyTEWXiixVzVBG6Q46zKLU9fjb`	**20 Medium**	Rapid pass-through only
T9	`TJDENsfBJs4RFETt1X1W8wMDc8M5XnJhCe`	**0 Low**	High-volume service: "Likely exchange or large service"
T10	`TV6MuMXfmLbBqPZvBHdwFsDnQeVfnmiuSi`	**0 Low**	Busy wallet with no risk indicators
T11	`TTjHn8qNVxazB3UroiekGRPuM2w2GwvNbm`	**Error**	Last character changed, so the checksum fails: "not a valid TRON address"

Also try: the **Fund flow trace** tab on T4 (backward, 1 hop). Funding wallets appear in the graph, and exchanges are shown in yellow.

2. Ethereum wallets

Investigate → **Ethereum wallets**.

#	Address	Expected	Why
E1	`0x8589427373D6D84E98730D7795D8f6f8731FDA16`	**100 High**	Tornado Cash donation address: sanctioned, frozen, mixer and exposure flags
E2	`0x098B716B8Aaf21512996dC57EB0615e2383E2f96`	**100 High**	Ronin bridge exploiter: sanctioned, frozen, pass-through
E3	`0x7F367cC41522cE07553e823bf3be79A889DEbe1B`	**100 High**	OFAC-listed address, frozen by the stablecoin issuer
E4	`0xd90e2f925DA726b50C4Ed8D0Fb90Ad053324F31b`	**100 High**	Tornado Cash Router (a contract): mixer, collection pattern. Has a **Contract audit** tab
E5	`0x98138788220cd08e17e364685d8af4edcea6ac0f`	**80 High**	Not sanctioned itself, but dealt directly with a mixer and with sanctioned addresses
E6	`0x7eee190785685c8c25d5cef5bb3204cfc725cdf3`	**35 Medium**	Pass-through, fan-in
E7	`0x27239549dd40e1d60f5b80b0c4196923745b1fd2`	**30 Medium**	Pass-through, possible P2P trader or service
E8	`0xfaa46f9a96a3d273e737de6e3b8a1444f4e81063`	**25 Medium**	Pass-through, mostly round amounts
E9	`0x28C6c06298d514Db089934071355E5743bf21d60`	**10 Low**	Binance hot wallet: "Likely exchange or large service"
E10	`0xd8dA6BF26964aF9D7eEd9e03E53415D37aA96045`	**0 Low**	Vitalik Buterin: no indicators (a clean control)
E11	`0x1234`	**Error**	Too short: "not a valid Ethereum address"

The same-key check runs on every address. None of these has an active TRON twin, so no "Same key on TRON" box appears.

3. Bitcoin addresses

Investigate → **Bitcoin addresses**.

#	Address	Expected	Why
B1	`125W5ek3DT6Zqy5S2iPt4FHQdNMCbZA3FU`	**80 High**	OFAC-listed, rapid pass-through

B2	`1648zkx5W1b55BZwCQnJ4byrbffUpTxfqQ`	**80 High**	Exposure to a sanctioned address, pass-through, CoinJoin mixing
B3	`123WBUDmSJv4GctdVEz6Qq6z8nXSKrJ4KX`	**75 High**	OFAC-listed, fan-in (collection)
B4	`392XxH946Q1BNcDuBtD3ZNdRF7vCKpnqYv`	**60 High**	Exposure to a sanctioned address, pass-through
B5	`17d1bsFWcG7rkQSWYHSQJPeSy4rijqjFWd`	**40 Medium**	Exposure to a sanctioned address only
B6	`bc1qnhsxlya64swh5mqyprk68yhaenxfhzkpng07uy`	**35 Medium**	Pass-through, fan-out (distribution)
B7	`34xp4vRoCGJym3xR7yCVPFHoCNxv4Twseo`	**20 Medium**	CoinJoin participation, listed with transaction links
B8	`bc1qm34lsc65zpw79lxes69zkqmk6ee3ewf0j77s3h`	**10 Low**	Very large exchange wallet: "Likely exchange"
B9	`bc1qm9gpkjf6h66uzxyhlctkwy6rzzc7f5avxqxpxd`	**0 Low**	No indicators
B10	`1KNm4K8GUK8sMoxc2Z3zU8Uv5FDVjrA72p`	**0 Low**	No indicators
B11	`bc1qnotanaddress`	**Error**	"That is not a valid Bitcoin address"

4. Cross-chain tracing

Investigate → **Cross-chain tracing**. Put each address in the box for its chain and press Run. Hop counts for exchange wallets change daily; the order of magnitude should hold.

To copy an address on one line, use the earlier tables: `TJDENs…` is T9, `TEEYCu…` is T4, `0x28C6…` is E9, `0x098B…` is E2, `0xd8dA…` is E10, `bc1qm34…` is B8, and `TTjHn8…` is the wallet in section 6.

#	TRON box	Ethereum box	Bitcoin box	Expected
X1	`TJDENsfBJs4RFETt1X1W8wMDc8M5XnJhCe`	`0x28C6c06298d514Db089934071355E5743bf21d60`	(empty)	**Many hops** (about 100, TRON → Ethereum), a mix of **High**, **Medium** and **Low** confidence. Two exchanges produce many coincidental matches, which is why hops are leads, not proof
X2	(empty)	`0x28C6c06298d514Db089934071355E5743bf21d60`	`bc1qm34lsc65zpw79lxes69zkqmk6ee3ewf0j77s3h`	**About 40 hops** (Bitcoin → Ethereum), mostly **High** confidence
X3	`TJDENsfBJs4RFETt1X1W8wMDc8M5XnJhCe`	(empty)	`bc1qm34lsc65zpw79lxes69zkqmk6ee3ewf0j77s3h`	**One hop or a few** (TRON → Bitcoin)
X4	`TJDENsfBJs4RFETt1X1W8wMDc8M5XnJhCe`	`0x28C6c06298d514Db089934071355E5743bf21d60`	`bc1qm34lsc65zpw79lxes69zkqmk6ee3ewf0j77s3h`	**All three chains**: about 125 hops in three directions
X5	`TTjHn8qNVxazB3UroiekGRPuM2w2GwvNbn`	`0x098B716B8Aaf21512996dC57EB0615e2383E2f96`	(empty)	**No hops**: two sanctioned wallets active in different years. Same-key check: twins not active

X6	`TEEYCuGDyeNkuDj4u6GQRXxXo3Nh29r2vP`	`0xd8dA6BF26964aF9D7eEd9e03E53415D37aA96045`	(empty)	**No hops**: unrelated wallets
X7	`TTjHn8qNVxazB3UroiekGRPuM2w2GwvNbn`	(empty)	(empty)	**Only the same-key check**: its Ethereum twin `0xc2d06dae…62f4` has no activity, and there are no hops because only one chain is given

Pick a case in "Case (optional)" before running, and the links are attached to that case and shown in its Case graph.

5. Smart-contract audit

Investigate → **Smart-contract audit**. Files are in `demo/contracts/` (upload tab). Contract scores: **High** ≥ 50, **Medium** 20–49, **Low** under 20 (High-impact finding 30 points, Medium 10, Low 2).

#	Input	Expected	Main findings
S1	`07_delegatecall_FibonacciBalance.sol`	**100 High**	arbitrary-send-eth, controlled-delegatecall, uninitialized-state
S2	`04_unchecked_call_KingOfTheEther.sol`	**100 High**	controlled-array-length, msg-value-in-nonpayable, unchecked-send
S3	`06_reentrancy_EtherBank.sol`	**60 High**	reentrancy-eth, msg-value-in-nonpayable
S4	`08_reentrancy_EtherStore.sol`	**32 Medium**	reentrancy-eth
S5	`01_reentrancy_SimpleDAO.sol`	**30 Medium**	reentrancy-eth (the flaw behind the 2016 DAO hack)
S6	`05_safe_Vault.sol`	**0 Low**	None: the control sample, written defensively
S7	`02_overflow_BECToken.sol`	**0 Low**	**Known miss**: the overflow is real, but Slither has no detector for it on Solidity 0.4
S8	`03_access_control_constructor.sol`	**0 Low**	**Known miss**: the misnamed constructor is not detected
S9	Deployed: `0xdAC17F958D2ee523a2206206994597C13D831ec7`	**100 High**	Tether (USDT), solc 0.4.18. Many Medium `erc20-interface` and Low findings add up. The score counts findings, not exploitability
S10	Deployed: `0xd90e2f925DA726b50C4Ed8D0Fb90Ad053324F31b`	**24 Medium**	Tornado Cash Router, solc 0.6.12, 39 files: unused-return. Turn off "Only the contract's own file" to see the library findings too

S9 and S10: use the "Deployed on Ethereum" tab, or the Contract audit tab on the Ethereum page. The compiler download and scan take up to a minute.

6. Mobile-money attribution

Investigate → **Mobile-money attribution** → "Import a case". In every test, enter this wallet and the rate **2700**, then upload the file from `demo/statements/`:

> **Trader's TRON wallet:** `TTjHn8qNVxazB3UroiekGRPuM2w2GwvNbn` · **USD/TZS rate:** 2700

The transfers are the wallet's real USDT transfers. The payments are made up and placed around them (`demo/make_statements.py`).

#	File	Expected	What it tests
M1	`mm1_high_repeated_trades.csv`	**1 High pair**	5 payments, each matching a transfer within minutes at a 1.5% spread: `TEST-ACC-0001`, 5 trades, evidence 3.35
M2	`mm2_medium_two_trades.csv`	**1 Medium pair**	2 matched trades: `TEST-ACC-0002`, evidence 1.53
M3	`mm3_low_single_trade.csv`	**1 Low pair**	One matched trade: `TEST-ACC-0003`. Tick **Low** in the Confidence filter to see it, because the default shows only High and Medium
M4	`mm4_no_match_timing.csv`	**No match**	Payments 3 hours before the transfers, outside the time window
M5	`mm5_no_match_spread.csv`	**No match**	Right times, but the amounts imply a 12% spread (the accepted range is up to 5%)
M6	`mm6_mixed_high_medium_low.csv`	**High** + **Medium** + **Low**	M1, M2 and M3 together plus 3 unrelated decoy payments: 8 of 11 rows match, and the decoys stay unmatched
M7	`statement_missing_column.csv`	**Error**	"Statement is missing columns: counterparty_account"
M8	`statement_bad_direction.csv`	**Error**	"direction must be 'in' or 'out' on every row"

The **Training sample** option needs no input. It shows the method on generated data.

7. Quick checks after testing

- **Entity graph**: search any address you looked up. It is now in the shared graph, with its links.
- **Audit and custody**: every lookup, trace, scan and import you did is listed. "Verify chain" should report the chain intact.
- **Cases**: "Open case" on any High result creates a case. Attach traced wallets as leads and decide them in the Review queue.